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Switching Transport Modes to Meet Voluntary Carbon Emission Targets

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The transport sector is the second-largest carbon emissions contributor in Europe, and its emissions continue to increase. Many producers are committing themselves to reducing transport emissions voluntarily, possibly in anticipation of increasing transport prices. In this paper we study a producer that has outsourced transport and has decided to cap its carbon emissions from outbound logistics for a group of customers. Setting an emission constraint for a group of customers allows for taking advantage of the portfolio effect. We focus on reducing emissions by switching transport modes within an existing network, because this has a large impact on emissions. In addition, the company sets the sales prices, which influence demand. The problem is solved by decomposing the multiproduct problem into several single-product problems, which we then analyze separately. Using the single-product solutions, we create an efficient frontier that reflects the trade-off between total carbon emissions and the total profit. It is observed that a diminishing rate of return applies in reducing emissions by switching transport modes. In a case study, we apply our method to a producer of bulk liquids and find that emissions can be reduced by 10% at only a 0.7% increase in total logistics cost.

Keywords: carbon emissions; green supply chains; sustainability; transport mode selection; Lagrange relaxation; pricing
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1. Introduction

In recent years, concern has been growing over the factors contributing to climate change. One of the important contributors to climate change is (freight) transport. In Europe, in 2006, around 23% of all carbon dioxide (CO_2) emissions were because of the transport sector, which makes it the second-biggest contributor after the energy sector (European Commission 2011). Between 30% and 40% of these emissions are because of freight transport (European Commission 2007). Two-thirds of freight transport emissions are attributable to road transport. It is estimated (European Commission 2009) that the energy demand for freight transport in 2030 will be 60% higher than its 1990 level (currently it is 36% higher), despite the increasing fuel efficiency of vehicles. In a study commissioned by the European Commission (EU Transport GHG 2011), it is estimated that innovations in vehicles and alternative fuels are not sufficient to meet the emission target levels for 2050, and additional economic and policy instruments are required. It is therefore important to investigate opportunities that reduce demand for high-energy transport modes.

In this article, we limit ourselves to carbon dioxide emissions and regulations, and abstain from other (mainly local) pollutants. Within Europe, emissions standards for vehicles (such as the Euro standards for trucks and fuel standards for barges) have mainly focused on local pollutants and on sulfur dioxide and nitrogen oxides emissions. The current concern for carbon dioxide and other greenhouse gases is mainly driven by a concern for global warming among policymakers. However, for most companies, there is an adjacent concern for an increase in prices of energy in general, and fossil fuels in particular. In this paper, we do not address the option of using alternative fuels (such as biodiesel or hybrid traction). For fossil-fueled transport types, there is a linear relation between fossil fuel consumption and carbon dioxide emissions. Therefore, the results presented in this paper can be directly related to fuel consumption. The exception for this direct relationship is the electric rail transport mode, because emissions for this mode are generated during electricity generation. In §5 we provide additional information on the fuel consumption.

In response to the growing carbon emissions from transport, policymakers have started developing emission regulations. In the European Union, carbon emission regulations for transport are currently limited to air transport, for which a separate emissions trading scheme (EU ETS) has been developed (European Commission 2012). In addition to emission regulation,

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it is observed that companies are also voluntarily reducing emissions, not only because of corporate responsibility, but also in an attempt to improve their market share, company image, and value. For example, the Carbon Disclosure Project (2011) reports that 294 of the Global 500 companies have voluntary emission reduction targets.

In this paper, we consider a company that is committed to reducing emissions from outsourced transport with a self-imposed emission target for the outbound transport emissions. All transport activities are outsourced to a third-party logistics provider (3PL). Transport emissions are a part of Scope 3 (indirect) emissions of the Greenhouse Gas Protocol (2011), unless company-owned vehicles are used. To achieve emission reductions, the company switches from high-carbon transport modes such as air or road, to low-carbon transport modes such as rail and water. The company determines indirectly which transport mode to use for each customer by specifying a maximum allowed transport time, since transport modes with lower emissions are generally also slower. Note that the choice of transport mode is one of the simplest emission abatement options in terms of implementation, provided that transport is outsourced. Moreover, it does not require a change in the supply chain, e.g., relocating warehouses. It is generally recognized that the transport mode used is one of the key factors that determines the emissions from transport. The potential gains in emission reductions are very large for modal shifts, e.g., the emissions (in kg CO_2 per tonne km) are around 0.01 for maritime shipping and 0.50 for shipping with a van (Defra 2010). Traditionally, the decision of which transport mode to use for which customer is based on the trade-off between the transport- and inventory-related costs. We extend this framework by bringing in carbon emissions as a third component. The problem of deciding which transport mode to use to ship products is an important determinant of transport emissions that needs to be considered in the light of meeting emission reduction targets.

In this paper we address the following question: how can the transport mode selection and sales prices be jointly optimized to meet an emission target efficiently for a group of customers? We are specifically interested in the portfolio effect, i.e., to compensate costly emission reductions on one lane with less costly reductions on another lane to achieve emission reductions at an overall lower cost.

We consider a profit-maximizing firm that is determining the sales price and the transport mode for each customer subject to an emission constraint for transport to all customers, where demand is price dependent. Observe that by optimizing the price, the company influences the demand for different products in such a way that the overall emission target is met. The assignment of transport modes to customers is typically a tactical decision; we therefore determine which mode maximizes the profit, based on the average demand figures. The logistics cost consists of a transport cost and a holding cost for the pipeline stock. We develop a solution procedure using Lagrangian relaxation. We note that our method is also applicable in a situation where the unit transport cost is a linear function of the emissions, allowing for the inclusion of a carbon cost, e.g., when transport prices increase as a consequence of emission regulation at the logistics service providers. We decompose the multiproduct problem into a number of single-product problems that are solved separately, which allows us to construct an efficient frontier of solutions representing the trade-off between profit and emissions. We find that the optimal sales price increases linearly and the profit decreases quadratically in the emission reduction target.

We apply our solution procedure to a business unit of Cargill, an international producer and marketer of food, agricultural, financial, and industrial products and services. We use data from 3PLs on transport cost and lead time, and we use price information given to us by Cargill. The transport emissions are calculated using a tool based on the NTM emission calculation methodology (NTM 2011). We find that intermodal transport, which is typically less carbon emitting, is more expensive (in terms of total logistics cost) than road transport for 63% of the customer lanes, implying that cost and emissions are not aligned (our experience with other companies showed that this observation is not unique to Cargill). We have observed that a diminishing rate of return applies when reducing emissions by switching transport modes. As a result, the solution is increasingly sensitive to the tightness of the emission target. We first solve a cost-minimization model that is more appropriate for the situation of Cargill, i.e., commodity goods. To not restrict our findings to the setting of Cargill, we also apply a profit-maximization model to investigate the impact of demand sensitivity with respect to price on the solutions. We observe that a 10% emission reduction can be achieved at only a 0.7% cost increase, compared to the minimum-cost solution, and it is possible to reduce emissions by 27%. The first 10% emission reductions correspond with a marginal cost of €100/tonne of CO_2. This suggests that a larger cost is required to reduce transport emissions compared to production emissions, which is in line with estimated carbon costs for transport in EU Transport GHG (2011). In a profit-maximization setting, we find that emissions can be reduced up to 30% at a 1.2% profit loss, which does not appear to be sensitive to price elasticity.

The main contribution of this research is threefold: First, we develop a model to achieve an overall

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emission target by optimizing transport modes and prices of products simultaneously. Second, we develop a solution procedure that takes into account an overall emission constraint for shipments to a *group of customers*. This allows us to exploit the portfolio effect, which implies that emissions are reduced on lanes for which this is less costly. The portfolio effect enables a certain emission target is met at a higher profit level compared to a situation with the same relative reduction target for each product. Our model with Lagrangian relaxation allows us to make conclusions about the price level required to obtain certain emission reduction levels. Lastly, we apply our analysis to a real-life case study and we develop several managerial insights.

The remainder of the paper is organized in the following structure: In §2 we position our work in the existing body of literature. The model and underlying assumptions are presented in §3. In §4 the analysis of the model is described. In §5, we apply the analysis to a real-life case study. We end with the conclusion in §6.

2. Related Literature

Parallel to the raising awareness of carbon emissions, the number of articles addressing emissions is steadily increasing. We first mention three articles that investigate the impact of transport emission regulation in Europe on transport costs: Abrell (2010) uses an economic general equilibrium model to determine the impact of several regulation scenarios on the welfare of individual (EU and non-EU) countries. He finds that exempting transport from emission regulation and shifting the reduction burden to other sectors leads to the smallest welfare reduction. Scheelhaase, Grimme, and Schaefer (2010) investigate the impact of the inclusion of aviation in the EU ETS on the competition of EU- and non-EU-based airlines. They find that non-EU-based airlines gain competitive advantage over EU-based airlines. In Cadarso et al. (2010), a method is developed to measure emissions from international freight transport and to allocate emissions based on consumer responsibility. According to the consumer responsibility principle, emissions are allocated to the country where goods are consumed, even when it is produced elsewhere. These models that study the impact of transport emission regulation have, in general, a high aggregation level, and as a result, do not explicitly model the decisions made by the producer(s).

Transportation research and inventory management are two major fields where the transport mode selection problem has been studied from various perspectives. In transport infrastructure planning models, the transport mode selection is the third step in the four-step transport planning model (see, for example, Johnston 2004; Rodrigue, Comtois, and Slack 2009). The other three steps involve trip generation, trip distribution, and route selection. Another stream of transportation literature focuses only on mode selection. In this field, the focus is a producer for which several available transport modes are compared based on several attributes, ranging from cost to lead time and accuracy, to derive which performs best in minimizing total logistics costs. A literature review of this field is given by Tyworth (1991). Also within this field, the transport emissions have been taken into account. In Blauwens et al. (2006), the effectiveness of policy measures that aim at moving away from road transport, because of congestion, to other transport modes is investigated. They take the perspective of a policymaker and investigate what policy is preferred, whereas we focus on a producer that aims at reducing its transport emissions. Two articles that study the transport mode selection decision including emissions for a specific case study without constructing a general model are Cholette and Venkat (2009) and Leal and de Almeida D'Agosto (2011). Cholette and Venkat (2009) determine for a wine supply chain the emissions from transport and warehousing and investigate the impact of different supply chain designs, including transport modes. Several delivery options are considered and evaluated in terms of costs and emissions. One of their findings is that the transport mode has a large impact on the total emissions of the supply chain. Leal and de Almeida D'Agosto (2011) consider the modal choice decision while taking into account socioenvironmental considerations for the case of shipping bioethanol for a fixed origin-destination pair.

Within inventory management/supply chain management, environmental impacts have been internalized by the work in the green supply chain management literature. Articles in this field deal with how much to produce or order from one or more sources, taking into account environmental impacts, which can range from (carbon) emissions to waste. The literature review by Srivastava (2007) provides an overview of the research on green supply chain management. Other extensive literature reviews on the topic of green supply chain management are: Corbett and Kleindorfer (2001a, b); Kleindorfer, Singhal, and Van Wassenhove (2005); Sasikumar and Kannan (2009).

One of the earlier works in the inventory management field that takes emissions into account is Penkuhn et al. (1997), which develops a production-planning model for the process industry with environmental constraints. In Hua, Cheng, and Wang (2011), the order quantity decision is reexamined, taking into account that emissions from transport are subject to an emissions-trading scheme. Our model

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considers a more complex setting with multiple products and transport modes and an overall emission constraint. We assume, moreover, that transport is outsourced for which variable emissions and order/transport costs are more appropriate. Yalabik and Fairchild (2011) determine how much to invest in abatement of production emissions under emission regulation, also in the presence of a competitor offering an identical product. In Hoen et al. (2014), a stochastic inventory model is extended to incorporate transport emissions. The transport mode and order-up-to level are jointly optimized in a single-product setting for a company subject to alternative emission regulations for transport. To the best of our knowledge, we are the first to consider a voluntary emission target for a group of items for which transport modes, from possibly more than two available, need to be selected.

To the best of our knowledge, we are the first to model the transport mode selection decision in a multi-item setting, including a pricing problem, and in which transport emissions are bound by a constraint. Further, we focus on the producer and we develop a formal model that takes into account cost and emissions from transport.

3. Model Description

We study a producer that ships products to customers. The producer wants to reduce the emissions related to transport of the products by shifting to a transport mode with lower carbon emissions and requests logistics service providers to provide per customer (origin-destination pair) the tonne-kilometer transport costs, the distance, and lead time for several (intermodal) transport modes. The problem to solve is which transport mode to use for shipments to each of the customers to meet a predetermined emission target for all customers together while maximizing the profit. Note that by varying the emission target, different solutions (in terms of average profit and total emissions) can be obtained.

The sales price offered to a customer should cover the cost for transportation; hence, the sales price can vary from customer to customer (e.g., in different markets). In this article, our focus is on tactical-level decision making, e.g., which transport mode to assign to a product-customer combination for the next year, rather than the operational-level decision making that takes account of short-term variations. We therefore consider the average demand to be the main factor in making the mode allocation and price-setting decisions. Accordingly, we build a deterministic model in which only the average demand (which is assumed to be linearly decreasing with the sales price) is taken into account, and not the higher moments. We also assume that the demand function is corrected for the impact of *production* emissions on demand. We consider a setting where the customers specify when products need to be delivered, giving the producer the flexibility on when to produce, provided that the goods are delivered on time.

We assume that the transport activity is outsourced and executed by a third-party logistics provider or a carrier. If transport would be executed by company-owned vehicles, a modal shift may in fact be a capital investment decision. Moreover, in that case, utilization of vehicles is very important, and few transport alternatives may exist. So, reducing emissions by shifting company-owned transport modes is not likely to be the most cost-efficient alternative. The company indicates per customer a maximum allowed transport time that indirectly determines the transport mode to use. The assumption that transport is outsourced has implications for our cost function and emissions structure: we only consider a variable component per unit shipped.

We assume that the producer voluntarily imposes a constraint on the emissions for the group of products, even if the transport emissions of the producer are not regulated. Self-imposed targets have been observed in practice and have been investigated by Reid and Toffel (2009) and Short and Toffel (2010). Companies may expect additional benefits by disclosing their emission targets and moving beyond environmental regulation, such as increased valuation of the firm or customer value, as studied by, among others, Klassen and McLaughlin (1996); Reinhardt (1999); Dowell, Hart, and Yeung (2000); Jacobs, Singhal, and Subramanian (2010).

Let $J = \{1, 2, \ldots, n\}$ $(j \in J)$ denote the set of customers to which units are to be shipped. In the remainder of this article we use the term "product" to refer to a combination of a customer (location) and a product type. Note that the model allows for multiple production locations but each product is supplied to a customer from one production location. For each unit of product j produced, the company incurs a unit cost, which is denoted by k_j $(k_j \geq 0)$. Note that this unit cost includes the direct labor and material cost incurred to manufacture one item. The weight of one unit of product j is denoted by w_j $(w_j > 0)$.

Let $I = \{1, 2, \ldots, m\}$ $(i \in I)$ denote the set of available transport modes. Please note that unless a truck is used to execute the transport, or the origin and destination are located near terminals, a combination of transport modes is required: intermodal rail or water transport. If it is undesirable that a particular mode is used for a customer, e.g., because of restrictions of the transportation network or the lead time; then the transport cost is set to infinite.

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For each unit of product j shipped with transport mode i, a transportation cost is incurred that is denoted by $c_{i,j}$. The transportation cost is a function of a cost factor t_i ($t_i \geq 0$), expressed as a monetary unit per tonne of cargo shipped over 1 kilometer (e.g., €/tonne km), the traveled distance, and the lead time. The distance traveled for product j also depends on the transport mode used and is denoted by $\delta_{i,j}$ ($\delta_{i,j} \geq 0$). The deterministic lead time of product j and transport mode i is denoted by $l_{i,j}$ ($l_{i,j} \geq 0$). The lead time is determined by the 3PL and includes waiting times between the different legs of a trip. The unit transportation cost associated with product j and mode i is then: $c_{i,j} = t_i \delta_{i,j} w_j$. We assume that the products are paid for by the customer when they are delivered. Hence, one unit of product j in transit corresponds with an opportunity cost of value k_j. Let $h_j = h_c k_j$ denote the holding cost associated with product j, where h_c ($h_c \geq 0$) denotes the opportunity cost (or holding cost) rate for the items in transit. The logistics cost associated with using transport mode i for one unit of product j is then: $u_{i,j} := c_{i,j} + h_j l_{i,j}$. Note that the parameter values of intermodal transport represent a weighted average for the total distance.

The emissions associated with transporting one unit of product j with mode i are denoted by $e_{i,j}$. We approximate $e_{i,j}$ with the following structure, which is based on emission measurement methodologies, e.g., NTM (2011):

$$e_{i,j} = w_j(a_i + b_i \delta_{i,j}),$$

where a_i and b_i are mode-specific emission constants. The fixed emission factor a_i ($a_i \geq 0$) is associated with the emissions generated during the beginning and end of a trip (most notably for air transport), and the variable emission factor b_i ($b_i > 0$) denotes the emissions generated per kilometer traveled. Both values are expressed per weight unit, e.g., kg, of load transported. Several methodologies are available to estimate the emission factors, such as Defra (2010) and NTM (2011).

In the remainder of this section, we describe our problem formulation in §3.1 and the Lagrangian relaxation of the problem formulation in §3.2.

3.1. Problem Formulation

We assume that the company sets the price and observes demand. Let p_j denote the sales price of product j when mode i is used and the demand (per time unit) is denoted by $q_j(p_j)$. Note that the sales price would differ when a different mode with its corresponding cost is used for product j. We use an additive demand function:

$$q_j(p_j) = Q_j - \epsilon_j p_j,$$

where Q_j ($Q_j \geq 0$) corresponds with the maximum demand and ϵ_j ($\epsilon_j \geq 0$) denotes the sensitivity of demand for product j to the sales price. To ensure that a nonnegative quantity is sold, we restrict p_j: $p_j \leq Q_j/\epsilon_j$.

The objective function is the profit per time unit, which is denoted by $\Pi_{i,j}(p_j)$, for product j and mode i and is determined by the profit per unit $(p_j - k_j - u_{i,j})$ and the quantity sold ($q_j(p_j)$):

$$\Pi_{i,j}(p_j) = -\epsilon_j p_j^2 + \big(\epsilon_j(u_{i,j} + k_j) + Q_j\big)p_j - (u_{i,j} + k_j)Q_j. \tag{1}$$

The total emissions per time unit (denoted by $\Gamma_{i,j}(p_j)$) for product j and mode i are determined by the unit emissions ($e_{i,j}$) and the quantity sold ($q_j(p_j)$):

$$\Gamma_{i,j}(p_j) = e_{i,j}(Q_j - \epsilon_j p_j). \tag{2}$$

We conclude this section with the definition of problem (P) in which one transport mode is selected for each product and the sales price is set to maximize profit under an overall emission constraint. The maximum allowed amount of overall carbon emissions is denoted by β ($\beta > 0$). Let $x_{i,j}$ ($x_{i,j} \in \{0,1\}$) denote whether mode i is used for product j. In addition, this implies that when a positive quantity is sold for product j, it is shipped with mode i. Further, we define the vectors $\mathbf{x}_j = (x_{1,j}, x_{2,j}, \ldots, x_{m,j})$ for $j \in J$ and $\mathbf{p} = (p_1, \ldots, p_n)$, $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)$.

$$\text{(P)} \qquad \max_{\mathbf{p}\in\mathbf{P},\, \mathbf{x}\in\mathbf{X}} \ \Pi(\mathbf{p},\mathbf{x}) = \sum_{j\in J}\sum_{i\in I} x_{i,j}\Pi_{i,j}(p_j)$$

$$\text{subject to} \ \ \Gamma(\mathbf{p},\mathbf{x}) = \sum_{j\in J}\sum_{i\in I} x_{i,j}\Gamma_{i,j}(p_j) \leq \beta,$$

where $\mathbf{P} = (\mathbf{P}_1, \ldots, \mathbf{P}_n)$, $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_n)$, $\mathbf{X}_j = \{\mathbf{x}_j \in \mathbb{R}^m \mid x_{i,j} \in \{0,1\}, \sum_{i\in I} x_{i,j} = 1\}$, and $\mathbf{P}_j = \{p_j \in \mathbb{R} \mid p_j \leq Q_j/\epsilon_j\}$, for $j \in J$. Note that $\Pi(\mathbf{p},\mathbf{x})$ and $\Gamma(\mathbf{p},\mathbf{x})$ are nonlinear, so it is a nonlinear problem. In this article, the focus extends beyond solving problem (P) and includes investigating when transport mode switches occur. For readers interested in efficiently solving problem (P), we present in Appendix A.1 an alternative formulation that results in a quadratic problem formulation.

Special case: *Cost-minimization model.* If the sales price and quantity are fixed for all products (on the medium-long term), then the problem reduces to selecting the mode for each product that minimizes costs given the overall emission constraint. Let q_j denote the sales quantity for product j, which is independent of the mode used. The revenue for product j is then fixed, and the only costs that are impacted by the transport mode decision are the logistics costs.

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Hence, the objective function is the total cost per time unit, denoted by $C_{i,j}$ for product j and mode i:

$$C_{i,j} = q_j(t_i\delta_{i,j}w_j + h_c k_j l_{i,j}), \tag{3}$$

and the corresponding emissions are $\Gamma_{i,j} = q_j e_{i,j}$. The problem formulation in the cost-minimization model follows directly from Equation (3) and problem (P).

3.2. Lagrangian Relaxation

Problem (P) is a special type of a knapsack problem, an assignment problem, i.e., for each product one transport mode is assigned (Fisher 1981). We decompose this problem into multiple single-product problems by Lagrangian relaxation. In Lagrangian relaxation, a penalty cost is introduced for violation of the constraint. The Lagrangian function for problem (P) is defined as

$$L(\mathbf{p},\mathbf{x},\lambda) = \sum_{j\in J}\left(\sum_{i\in I} x_{i,j}\Pi_{i,j}(p_j)\right) - \lambda\left(\sum_{j\in J}\left(\sum_{i\in I} x_{i,j}\Gamma_{i,j}(p_j)\right) - \beta\right), \tag{4}$$

where $\lambda \geq 0$ is the Lagrangian multiplier. As λ increases, it reduces the emissions by charging a higher penalty. Since the profit and emission function are separable in j, as are the implicit constraints $((\mathbf{p},\mathbf{x}) \in (\mathbf{P},\mathbf{X}))$, the Lagrangian is also. Hence, we rewrite the Lagrangian as:

$$L(\mathbf{p},\mathbf{x},\lambda) = \sum_{j\in J} L_j(p_j,\mathbf{x}_j,\lambda) + \lambda\beta, \tag{5}$$

where

$$L_j(p_j,\mathbf{x}_j,\lambda) = \sum_{i\in I} x_{i,j}\Pi_{i,j}(p_j) - \lambda\sum_{i\in I} x_{i,j}\Gamma_{i,j}(p_j) \tag{6}$$

is the *decentralized Lagrangian* for product j. Note that the Lagrangians are only connected by a single multiplier λ of the emission constraint.

For a given value of λ, let $(p_j^*(\lambda), \mathbf{x}_j^*(\lambda))$ denote the solution that maximizes the decentralized Lagrangian of product j over $(p_j,\mathbf{x}_j) \in (\mathbf{P}_j,\mathbf{X}_j)$. Then the solution $(\mathbf{p}^*(\lambda),\mathbf{x}^*(\lambda))$, where $\mathbf{x}^*(\lambda) = (\mathbf{x}_1^*(\lambda),\ldots,\mathbf{x}_n^*(\lambda))$ maximizes the Lagrangian over $(\mathbf{p},\mathbf{x}) \in (\mathbf{P},\mathbf{X})$ for that value of λ.

If $\beta = \Gamma(\mathbf{p}^*(\lambda),\mathbf{x}^*(\lambda))$, then, by the *Everett result* (Everett 1963), $(\mathbf{p}^*(\lambda),\mathbf{x}^*(\lambda))$ is the optimal solution to problem (P) and the constraint will be met with equality. By varying the value of λ, we obtain different optimal solutions to problem (P) for specific values of β. It follows from Theorem 1 in Fox (1966) that these solutions are efficient solutions for the multicriteria problem (Q) (without constraints):

$$\text{(Q)} \quad \max_{\mathbf{p}\in\mathbf{P},\,\mathbf{x}\in\mathbf{X}} \Pi(\mathbf{p},\mathbf{x}) \quad \min_{\mathbf{p}\in\mathbf{P},\,\mathbf{x}\in\mathbf{X}} \Gamma(\mathbf{p},\mathbf{x}).$$

Solving problem (Q) results in a set of solutions from which one can observe the static impact of the emission target. A decision maker can then pick a favorable solution from the set of efficient solutions.

The decentralized Lagrangian can be separated further in mode i, because it is separable and only connected by the implicit constraint $(p_j,\mathbf{x}_j) \in (P_j,\mathbf{X}_j)$. We denote this function by $L_{i,j}(p_j,\lambda)$: $L_{i,j}(p_j,\lambda) = \Pi_{i,j}(p_j) - \lambda\Gamma_{i,j}(p_j)$. The decentralized Lagrangian can then be rewritten as follows, which follows directly from Equation (6):

$$L_j(p_j,\mathbf{x}_j,\lambda) = \sum_{i\in I} x_{i,j}L_{i,j}(p_j,\lambda). \tag{7}$$

Solving the Lagrangian relaxation of problem (P) for a specific λ is then done in time proportional to nm by evaluating $\max_{i\in I,\,p_j\leq Q_j/\epsilon_j} L_{i,j}(p_j,\lambda)$ for each j and setting the associated $x_{i,j} = 1$ and the remaining $x_{i,j}$ to zero (Fisher 1981).

4. Analysis

In this section we describe the solution procedure to determine the optimal solution of the Lagrangian relaxation of problem (P). First, we describe the solution for a fixed value of the emission penalty cost (λ) in §4.1. For each of the products, we first solve the pricing problem and then solve the assignment problem. The solutions for each of the products straightforwardly determine the solution to the Lagrangian relaxation of problem (P). In §4.2, we study the efficient solutions and profit-emissions trade-off for a single product and, in an example, for a two-product situation. Lastly, we describe in §4.3 how the product characteristics impact which mode is selected and how the total set of available transport modes can be reduced to include only modes that are selected for some range of the emission target.

4.1. The Fixed-Emission Target Problem

The pricing problem. Function $L_{i,j}(p_j,\lambda)$ is derived from Equations (1) and (2):

$$L_{i,j}(p_j,\lambda) = \epsilon_j\left(-(p_j)^2 + p_j\left(z_{i,j}(\lambda) + k_j + \frac{Q_j}{\epsilon_j}\right)\right) - Q_j(z_{i,j}(\lambda) + k_j), \tag{8}$$

where $z_{i,j}(\lambda) := u_{i,j} + \lambda e_{i,j}$, which represents the logistics cost, including emissions. We note that this function can also represent a situation in which an emission cost (λ) applies to transport. It is observed that $L_{i,j}(p_j,\lambda)$ is concave in p_j. Let $p_{i,j}^*(\lambda)$ maximize $L_{i,j}(p_j,\lambda)$; then,

$$p_{i,j}^*(\lambda) = \min\left\{\frac{1}{2}\left(z_{i,j}(\lambda) + k_j + \frac{Q_j}{\epsilon_j}\right), \frac{Q_j}{\epsilon_j}\right\},$$
$$q_j(p_{i,j}^*(\lambda)) = \max\left\{\frac{1}{2}(Q_j - \epsilon_j(z_{i,j}(\lambda) + k_j)), 0\right\}. \tag{9}$$

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The optimal sales price (sales quantity) is increasing (decreasing) in h_c, k_j, $l_{i,j}$, t_i, $\delta_{i,j}$, and w_j. The optimal sales price and quantity are decreasing as a function of ϵ_j and increasing as a function of Q_j. The corresponding maximum value is denoted by $L^*_{i,j}(\lambda) := L_{i,j}(p^*_{i,j}(\lambda), \lambda)$:

$$L^*_{i,j}(\lambda) = \begin{cases} \frac{1}{4}\left(\epsilon_j(z_{i,j}(\lambda)+k_j)^2 - 2Q_j(z_{i,j}(\lambda)+k_j) + \frac{Q_j^2}{\epsilon_j}\right) \\ \qquad \text{if } z_{i,j}(\lambda) \le \frac{Q_j}{\epsilon_j} - k_j, \\ 0 \qquad \text{otherwise.} \end{cases} \tag{10}$$

We note that $L^*_{i,j}(\lambda)$ is decreasing for $z_{i,j}(\lambda) \le Q_j/\epsilon_j - k_j$ and constant (at 0), otherwise. Thus, the decentralized Lagrangian is quadratic in λ. From the expressions, it can be seen that a tighter emission constraint, i.e., an increase of λ, results in an increase in the optimal price and a decrease in optimal demand, profit, and emissions. This is caused by the fact that an emission reduction for product j and mode i is realized by selling fewer products at a higher price.

The assignment problem. We next describe the solution that maximizes the decentralized Lagrangian of product j: determine the mode that maximizes $L_j(p_j, \mathbf{x}_j, \lambda)$ given the value of λ and the optimal sales prices $p^*_{i,j}(\lambda)$. For a given λ, only the value of $z_{i,j}(\lambda)$ determines the differences in the values of $L^*_{i,j}(\lambda)$ for different modes of product j. This implies that if the values of $z_{i,j}(\lambda)$ for the two modes are equal, then we are indifferent between selecting a product with lower logistics costs and higher emissions, and higher logistics cost and lower emissions. Consider two modes: $y_1, y_2 \in I$. Using the convexity of Equation (10), we find that $z_{y_1,j}(\lambda) \le z_{y_2,j}(\lambda) \iff L^*_{y_1,j}(\lambda) \ge L^*_{y_2,j}(\lambda)$ for $0 \le z_{y_1,j}(\lambda), z_{y_2,j}(\lambda) \le Q_j/\epsilon_j - k_j$, which leads to the following proposition.

Proposition 1. *Consider product $j \in J$ and $\lambda \ge 0$. Mode y maximizes the decentralized Lagrangian, $L^*_j(\lambda) = L^*_{y,j}(\lambda)$ and $x_{y,j} = 1$ for $y = \arg\min(z_{i,j}(\lambda)\colon z_{i,j}(\lambda) > 0, i \in I)$.*

Proof. Proof in Appendix A.2.

So, for product j the transport mode that minimizes total logistics cost, including emission cost, is selected. It can be concluded from Proposition 1 that a mode is not selected if another mode has both lower logistics cost and unit emissions. By simply combining the selected mode and corresponding optimal sales price for each of the products, one finds the solution to the Lagrangian relaxation of problem (P).

We note that allowing multiple modes to be used per product would not affect the optimal solution compared to our model, because only one mode is selected for a given value of λ, as long as the demand is deterministic.

Special case: cost minimization. In the cost-minimization model the decentralized Lagrangian follows from Equations (3) and (7):

$$L'_j(\mathbf{x}_j, \lambda) = \sum_{i \in I} x_{i,j} q_j (u_{i,j} + \lambda e_{i,j}).$$

Minimizing the decentralized Lagrangian over $\mathbf{x}_j \in \mathbf{X}_j$ is equivalent to selecting the mode that minimizes $z_{i,j}(\lambda)$, and the results for the profit-maximization model apply.

4.2. The Efficient Solutions for Varying Emission Targets

In this section, we examine the trade-off between profit, emissions, and the optimal sales price. We measure the impact by changing the value of λ ($\lambda \ge 0$). Consider modes i, y_1, $y_2 \in I$, and $u_{y_1,j} < u_{y_2,j}$. If mode y_2 is preferred over mode y_1 for any positive value of λ, then it is additionally required that $e_{y_1,j} > e_{y_2,j}$. If this holds, then let $\tilde{\lambda} \ge 0$ denote the value of λ such that $z_{y_1,j}(\tilde{\lambda}) = z_{y_2,j}(\tilde{\lambda})$, i.e., mode y_1 is selected for $\lambda \le \tilde{\lambda}$ and mode y_2 for $\lambda \ge \tilde{\lambda}$.

The total emissions associated with $q_{i,j}(p^*_{i,j}(\lambda))$, mode i, and product j are denoted by $\Gamma^*_{i,j}(\lambda)$:

$$\Gamma^*_{i,j}(\lambda) = \tfrac{1}{2} e_{i,j}\big(Q_j - \epsilon_j(u_{i,j} + \lambda e_{i,j} + k_j)\big). \tag{11}$$

Now let $\Gamma^*_j(\lambda) = \Gamma^*_{i,j}(\lambda)$ when mode i is selected. The switch from mode y_1 to y_2 leads to a strict decrease of emissions ($\Gamma^*_{y_1,j}(\tilde{\lambda}) > \Gamma^*_{y_2,j}(\tilde{\lambda})$), which can be observed from Equation (11) and the fact that $e_{y_1,j} > e_{y_2,j}$. This implies that not for every value of the emissions Γ_j, say α, there exists a value of λ such that $\alpha = \Gamma^*_j(\lambda)$.

The optimal sales price for product j and mode i, as given in Equation (9), can be rewritten as a function of the optimal emissions ($\Gamma^*_{i,j}$) denoted by α (by inserting $\lambda = (\Gamma^*_{ij})^{-1}(\alpha)$):

$$p^*_{i,j}(\alpha) = Q_j - \frac{\alpha}{e_{i,j}}.$$

It follows that for a given product and mode, $p^*_{i,j}$ is linearly decreasing in α (increasing in emission reductions), i.e., the price is highest when the emissions are 0. This implies that the rate of increase is increasing when switching from mode y_1 to y_2, since $e_{y_1,j} > e_{y_2,j}$. Moreover, $p^*_{y_1,j}(\tilde{\lambda}) = p^*_{y_2,j}(\tilde{\lambda})$, i.e., the price increases continuously.

The realized profit associated with $p^*_{i,j}(\lambda)$, mode i, and product j is denoted by $\Pi^*_{i,j}(\lambda)$:

$$\Pi^*_{i,j}(\lambda) = \frac{1}{4}\left(\frac{1}{\epsilon_j}\big(Q_j - \epsilon_j(u_{i,j} + k_j)\big)^2 - \epsilon_j(e_{i,j}\lambda)^2\right).$$

It is observed that $\Pi^*_{y_1,j}(\tilde{\lambda}) > \Pi^*_{y_2,j}(\tilde{\lambda})$, since $e_{y_1,j} > e_{y_2,j}$. Hence, a switch from mode y_1 to y_2 results

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in a decrease in profit. Let the total emissions ($\Gamma^*_{i,j}$) be α; then

$$\Pi^*_{i,j}(\alpha) = \frac{1}{e_{i,j}}\left(\alpha\left(\frac{Q_j}{\epsilon_j} - (u_{i,j} + k_j)\right) - \frac{1}{\epsilon_j e_{i,j}}\alpha^2\right).$$

From this equation, it follows that $\Pi^*_{i,j}$ is quadratic and increasing in α (decreasing in emission reductions) for a given product and mode. So there is a diminishing rate of return, i.e., emission reductions become increasingly costly. This also implies that for low values of emission reduction, the solution (in terms of profit and sales price) is relatively insensitive, but as the constraint tightens (the reduction increases) the solution becomes more sensitive. Note that this result also holds for the solutions to problem (Q), since they follow from addition of the solutions for each of the products. We next present a small theoretical example to illustrate the results of this section and focus on the impact of the emission target on the results. In §5 we apply this model to a case study.

Example. Consider $J = \{a, b\}$ and $I = \{1, 2, 3, 4, 5, 6\}$, and the following parameter values: $Q_a = 100$, $Q_b = 80$, $\epsilon_a = 1.25$, $\epsilon_b = 1.10$, $k_a = 15$, $k_b = 6$. The logistics cost and unit emissions for each mode and product are given in Table 1.

We determine for each mode and each product the sales price that maximizes the Lagrangian ($L^*_{i,j}(\lambda)$) as we vary the value of λ. Then we select the mode that maximizes the value of $L^*_{i,j}(\lambda)$ over all modes to find the value of $L^*_j(\lambda)$. We observe that for each product some modes are not selected for any value of λ: for product a modes 2 and 4 and for product b modes 2, 3, and 5. Next, we determine the maximum profit, sales price as a function of the emission reduction relative to the solution for $\lambda = 0$, which is displayed in Figure 1.

As described previously, a switch from one mode to another for a product results in a change in profit and total emissions, which is represented by the "gaps" in the figures. As described before, it can be seen that the sales price is piecewise linearly increasing in the emission reduction for each mode, and the profit is quadratically decreasing where they are defined.

The efficient solutions to problem (Q) for the example are displayed in Figure 2. The solutions are found by simply adding the total profit and emissions for products a and b. The solutions are distinguished for the seven different combinations (in terms of modes selected for each product). The notation "6*" refers to the fact that no units of product b are sold. Note also that the combined profit for both products is quadratic in the total emissions reduction where it is defined. As a result, the solution is increasingly

Table 1 Parameter Values for Logistics Cost and Unit Emissions

Mode i	1	2	3	4	5	6
$e_{i,a}$	1.00	0.80	0.60	0.55	0.25	0.10
$u_{i,a}$	5	10	13	20	30	50
$e_{i,b}$	2.00	1.90	1.91	1.25	1.20	0.90
$u_{i,b}$	10	12	15	20	21	25

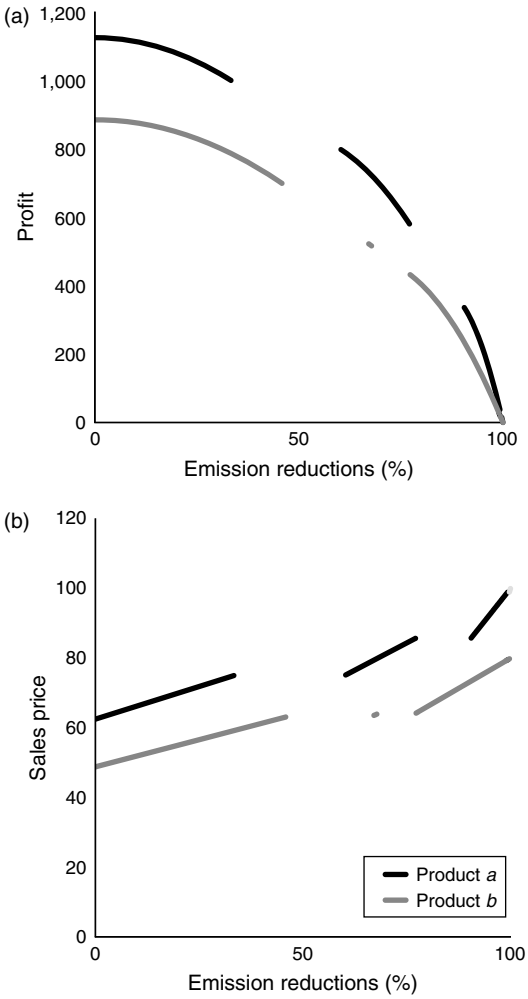


Figure 1 Profit (a) and Sales Price (b) as a Function of Emission Reductions

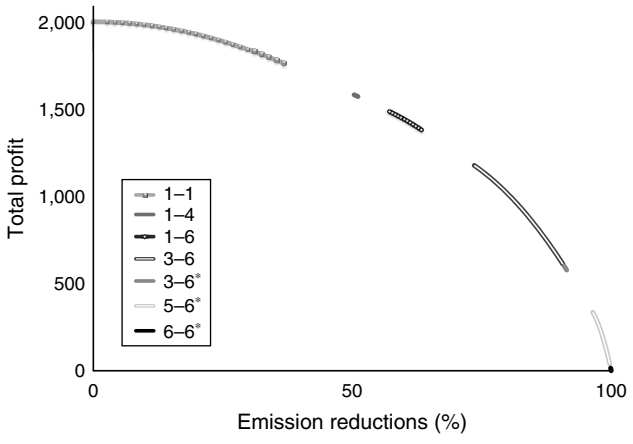


Figure 2 Solutions to Problem (Q)

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sensitive to the emission reduction. It is expected that the gaps decrease when the number of products increases, since each single product contributes a smaller part of the total emissions.

4.3. Results on Mode Selection

In this section we derive conditions that ensure that a mode is selected for at least one emission target. Consider two modes i_1 and i_2 ($i_1, i_2 \in I$), and assume without loss of generality $u_{i_1,j} \le u_{i_2,j}$. We define $\lambda^j_{i_1,i_2}$ such that $L^*_{i_1,j}(\lambda^j_{i_1,i_2}) = L^*_{i_2,j}(\lambda^j_{i_1,i_2})$ (and $z_{i_1,j}(\lambda^j_{i_1,i_2}) = z_{i_2,j}(\lambda^j_{i_1,i_2})$), then:

$$\lambda^j_{i_1,i_2} := \frac{u_{i_1,j} - u_{i_2,j}}{e_{i_2,j} - e_{i_1,j}} = \frac{h_c k_j (l_{i_1,j} - l_{i_2,j}) + w_j (t_{i_1}\delta_{i_1,j} - t_{i_2}\delta_{i_2,j})}{w_j (a_{i_2} - a_{i_1} + b_{i_2}\delta_{i_2,j} - b_{i_1}\delta_{i_1,j})}. \quad (12)$$

Equation (12) represents a threshold value of λ such that mode i_1 is preferred over mode i_2 for λ values less than the threshold value, or vice versa. As before, note that $\lambda^j_{i_1,i_2} \ge 0$ only if $e_{i_1,j} > e_{i_2,j}$. The expression for $\lambda^j_{i_1,i_2}$ provides us with a number of insights that we state in the following.

Observation. Consider two available transport modes for product j: $i_1, i_2 \in I$ and $u_{i_1,j} \le u_{i_2,j}$, and $e_{i_1,j} > e_{i_2,j}$. Using Equation (12), we derive the following results:

(a) As the weight of product j increases, the value of $\lambda^j_{i_1,i_2}$ decreases, i.e., for heavier products a switch to mode i_2 occurs for a smaller emission reduction target (for smaller values of λ). So, for heavier products, cleaner and more expensive modes are used for smaller emission reductions.

(b) As the value of product j increases, the value of $\lambda^j_{i_1,i_2}$ increases, i.e., for more valuable products a switch to mode i_2 occurs for a larger emission reduction target. So, for more valuable products pipeline inventory is more expensive, favoring a mode with shorter lead time.

(c) If $l_{i_1,j} \le l_{i_2,j}$, then an increase in $l_{i_1,j}$ decreases $\lambda^j_{i_1,i_2}$. A longer lead time results in more pipeline inventory and relatively more costs, compared to mode i_2. A similar observation is made for an increase in t_{i_1} if $t_{i_1,j} \le t_{i_2,j}$.

(d) If the unit emissions are similar for modes i_1 and i_2, then $\lambda^j_{i_1,i_2}$ increases. This implies that if the savings in emissions are marginal, a switch to mode i_1 is not made until profits have decreased to a larger extent.

For product j, the total available set of transport modes can be reduced to only those modes that are selected for at least one value of λ, which allows for faster execution of the solution procedure. This is formalized by two conditions in the following theorem.

Theorem 1. *Consider $i_2 \in I$ and any $i_1, i_3 \in I$ such that $u_{i_1,j} < u_{i_2,j} < u_{i_3,j}$ and $e_{i_1,j} > e_{i_2,j} > e_{i_3,j}$. Transport mode i_2 is selected for at least one value of λ if it meets conditions* (a) *and* (b) *for any i_1, i_3 that are selected*:

Condition (a) $e_{i_2,j} \le \tilde{e}^j_{i_1,i_2}$.

Condition (b) $e_{i_2,j} \le \bar{e}^j_{i_2}(i_1, i_3)$, *where*

$$\tilde{e}^j_{i_1,i_2} = e_{i_1,j}\frac{\hat{D}_j/\epsilon_j - k_j - u_{i_2,j}}{\hat{D}_j/\epsilon_j - k_j - u_{i_1,j}}, \qquad \bar{e}^j_{i_2}(i_1, i_3) = e_{i_1,j} + (e_{i_3,j} - e_{i_1,j})\frac{u_{i_1,j} - u_{i_2,j}}{u_{i_1,j} - u_{i_3,j}}. \quad (13)$$

Proof. See Appendix A.2.

We note that the mode that minimizes the total logistics cost for product j is preferred for small values of λ, i.e., at least for $\lambda = 0$. Condition (a) in the theorem ensures that Equation (12) represents a value in the range of λ for which L^*_{ij} is positive. Condition (b) follows from interpolation of modes i_1 and i_3, and requires that all selected modes form a convex hull (in terms of u_{ij} and e_{ij}). An increase in one of the parameters may ensure that one of the conditions is not met and that that particular mode is not selected.

Special case: *Cost minimization.* The results presented in Theorem 1 also hold for the cost-minimization case with a small exception. Condition (a) is less strict for the cost-minimization model: it is only required that $u_{i_1,j} \le u_{i_2,j}$ and $e_{i_1,j} > e_{i_2,j}$.

5. Case Study

In this section we apply our method to a real-life case study. The case applies to the products of a particular business unit of Cargill that are produced and shipped to customers in Europe. The products are food ingredients that are supplied to the customers in dedicated containers. Cargill decided to cap the emissions from transport by shifting away from road (or ferry) transport to intermodal transport. In a request for quotation (RFQ), third-party logistics providers were asked to provide Cargill with intermodal bids, which are used in our analysis. The emissions were calculated using the NTM methodology, which is described in §5.1. The market where Cargill operates is typically agricultural commodity goods in which limited flexibility exists for sales prices. Hence, we apply the cost-minimization model to the case study data in §5.2. Nevertheless, we still apply a profit-maximization model in §5.3 to not confine our study to the setting of Cargill, and investigate for other products the impact of demand sensitivity on emission reductions.

5.1. Emission Calculation

To obtain the value of carbon emissions associated with the transportation of a single unit of product,

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two steps have to be taken. First, one needs to obtain information on the carbon emissions for the entire vehicle. Second, an allocation of emissions to a single product has to be made. If the fuel consumption of the vehicle is known, then the first step is straightforward. For outsourced transportation, however, information on the actual fuel consumption is generally not shared by the transport service provider, and hence, the estimate needs to be made based on the specific activity (e.g., transport of weight x over distance y) and general information on the type of vehicle used. The second step always involves a certain degree of approximation unless a number of identical products are shipped.

In this paper, we use an activity-based method to determine the carbon emissions associated with the vehicle, because we deal with outsourced transportation. We have used the NTM methodology, because it focuses on Europe (which is where our data apply), allows for a relevant level of detail, and provides parameter estimates (NTM 2011). For further information on activity-based carbon emission estimation models and the consequences of inappropriate aggregation, we refer to Vélázquez-Martínez et al. (2014).

The first step in deriving the carbon emissions is to specify the modality and vehicle type. Using the NTM methodology, we can derive the emission factors associated with transporting the maximum load. In general, two emission factors are used: a fixed and a variable emission factor. The variable emission factor specifies the emissions associated with traveling a distance of 1 km. The fixed emission factor corresponds with the start and end of a transport leg, irrespective of the distance, i.e., taking off and landing for an airplane.

Allocation of emissions to a single unit of the product is done based on the weight of the product: 3PLs use the volumetric weight of a product for their transport cost to account for the fact that for low-density products, the volume of the product is restrictive, in contrast to the weight for heavier products. For the products, we consider the weight to be restrictive, so we allocate emissions based on the weight. The general structure for the unit emissions is:

$$e_{i,j} = (A_i + B_i \delta_{i,j}) \frac{w_j}{\tilde{w}_i},$$

where A_i is the constant emission factor for the vehicle (in tonne CO_2), B_i is the variable emission factor for the vehicle (in tonne CO_2/km), and $\tilde{w}_i$ is the average load of the vehicle (in tonne).

Five transport modality types are considered in the case study: road transport, intermodal rail transport, intermodal water transport (coastal shipping or short sea), ferry transport (road transport plus a ferry crossing), and inland water transport (using rivers and/or canals). Note that ocean and air shipping are not feasible transport modes for the case study setting. For the remainder of the article, we denote these modes by road, rail, short sea, ferry, and inland water. We have used the emission factors and assumptions specified in NTM Rail (2008), NTM Road (2008), and NTM Water (2008). In Appendix A.3 the required parameters and assumptions for the emission calculation are specified.

Table 2 Emissions per Tonne Km for the Case Study

Modality	Average	Minimum	Maximum
Road	47.64	46.37	50.51
Ferry	49.99	47.70	54.94
Rail	23.01	10.01	35.12
Inland water	24.56	21.91	26.01
Short sea	16.80	7.74	39.38

Table 2 contains information on the variable emission factor in g CO_2/tonne km for the data set per modality type. In general, it can be observed that road transport (and ferry) result in the highest carbon emissions, then rail, and lastly short sea transport.

5.2. Cost-Minimization Model

Data obtained from the 3PLs are the *payload* (the maximum load of a shipment), the modality type, the vehicle/vessel type, the loading and unloading location (location of the intermodal terminal), transport cost, and lead time. The annual demand and the unit cost of the product (obtained from Cargill), and emissions per tonne of cargo per product-mode combination, were calculated. The company always ships full containers; therefore, the number of shipments is determined by the payload. In total, the data set contains 56 products, of which the origins and destinations are all located within Europe. The 3PLs made 279 intermodal bids, and the amount of bids per product varies from 0 to 14 (no intermodal bids were received for five products). Multiple bids of the same modality type, e.g., rail, for one product are allowed as long as they differ in terms of lead time, transport cost, or emissions. In total, 335 product-mode combinations are available, including the current modality: road or ferry for each product.

The annual demand (q_j) varied between 300 and 6,000 tonne (on average 1,500), which corresponds with 11–240 shipments per product per year (payload between 21 and 29 tonne). The lead time ($l_{i,j}$) is between 1 and 12 days. The transport cost per shipment ($c_{i,j}$), which takes into account the distance of the route and the weight of the shipment, expressed in normalized monetary units varies between 1 and 6.2 (on average 1.9). The distance of the product ($\delta_{i,j}$) varies from 300 km to 3,300 km (on average 1,150 km). A holding cost rate of 25% per year was assumed.

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Applying the solution procedure while varying the Lagrangian parameter has resulted in a set of efficient solutions that are displayed in Figure 3. The solutions are expressed in terms of cost increase and emission reduction compared to minimum cost solution ($\lambda = 0$). Figure 3 also displays the cost per tonne of unit emissions as a function of the emission reductions. Note that the vertical axis Figure 3(b) has a logarithmic scale.

When comparing the cost-minimizing solution to the actual setting applied in practice, we observe that some cost reductions are possible that also decrease emissions. For our analysis, we assume that the cost-minimizing solution is applied in the absence of emission targets. It can be seen that the costs increase exponentially, i.e., a diminishing rate of return, which is in line with our finding in §4.2. As a result, the curve is relatively flat for the first 10% emission reduction. A maximum emission reduction of 27% can be achieved, at a cost increase of 30%. Given the size of the total cost over all products (several million euros), this is substantial. The total emissions are in the order of several thousand tonnes per year. In this case, the company can reduce emissions by 10%, virtually without a cost increase (0.7%).

The first 10% emission reductions correspond with a marginal cost of €100/tonne of CO_2. This value is higher than current prices of allowances in the EU ETS (around €15/tonne). This is in line with a finding in the EU Transport GHG (2011) study that a high CO_2 tax is expected to substantially reduce emissions from transport, whereas a lower cost has very limited impact: €180/tonne results in 20% emission reductions in 2050, compared to 1990 levels (in contrast to only 1.4% reduction for €20/tonne).

In addition, we determined the share of emissions attributable to each of the modality types for the solutions, represented by Figure 4. Observe that inland water is never selected because other modes result in lower emissions. Moreover, in Table 3, the emissions per modality type are compared for the minimum-cost transport allocation and the minimum-emission transport allocation. All emissions are expressed as a share of the emissions of the minimum-cost solution.

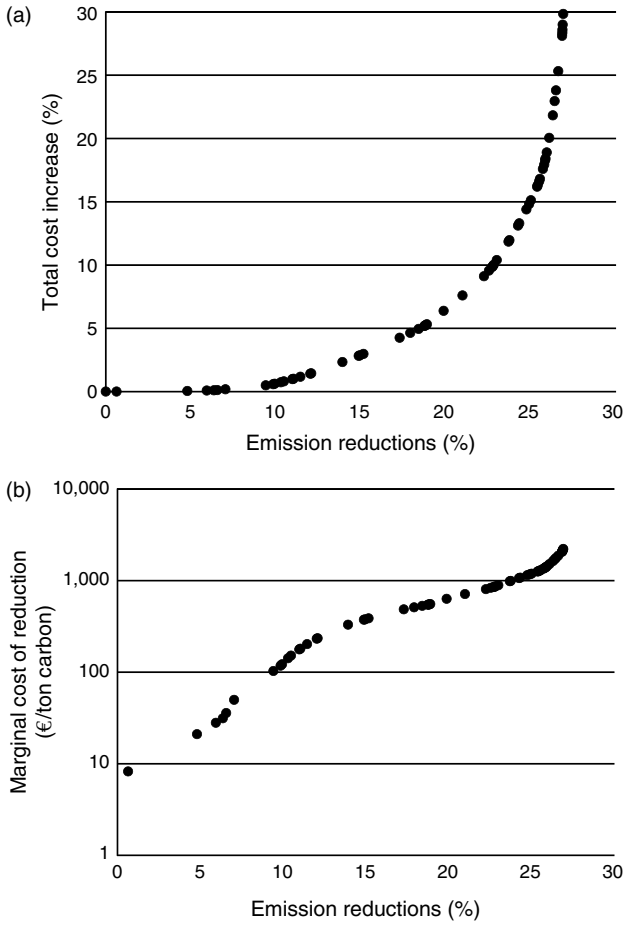


Figure 3 **Relative Cost Increase (a) and Marginal Emission Reduction Cost (b) as a Function of Emission Reductions**

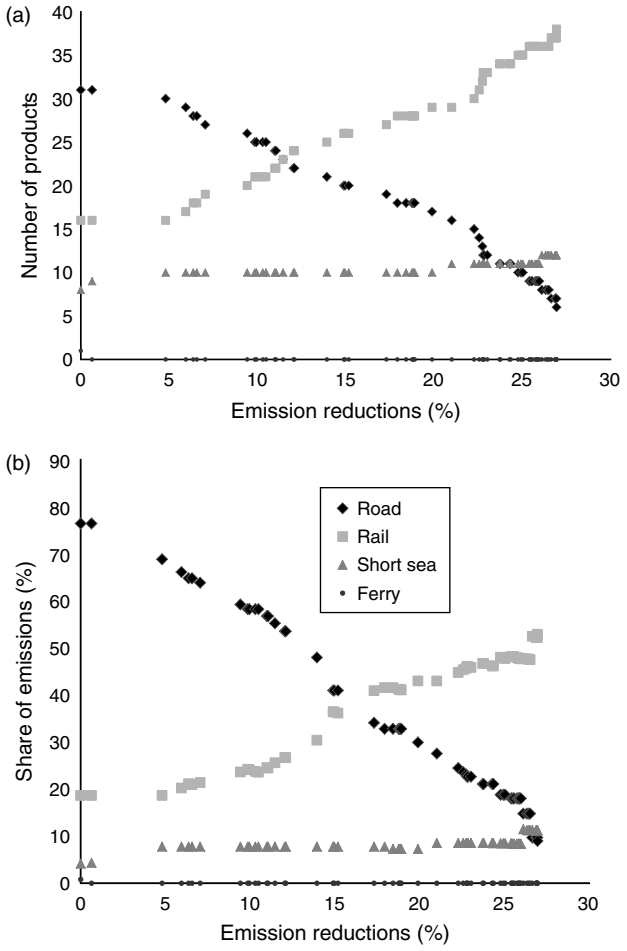


Figure 4 **The Share of the Emission per Modality Types as a Function of the Emission Reduction, Both in Number of Products (a) and Fraction of Emissions (b)**

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Table 3 Transport Mode Shares

	Cost-optimal transport assignment		Emission-optimal transport assignment	
	No. of products	Emission share (%)	No. of products	Emission share (%)
Road	31	76	6	9
Rail	16	19	38	53
Short sea	8	4	12	11
Ferry	1	1	0	0
Inland water	0	0	0	0

In the minimum-cost solution, the majority of emissions are because of road transport, and rail transport is the second-largest contributor. This is reduced to less than 10% in the emission-optimal solution, mainly because of a shift toward rail transport. Also note that a switch is made from ferry transport to short sea transport for one product. It can be concluded that for this case study, switching from road to rail transport provides the largest emission reduction potential.

From this case study we obtain the following insights: First, in contrast with the general belief, intermodal transport is not necessarily the most cost-efficient option in meeting emission targets. In particular, we observe that for a number of products (31), road transport results in lowest total logistics cost, which implies that for those particular products, although intermodal might result in lower emissions than road transport, it is also more expensive. The cost difference is because of longer lead times and additional handling costs. We believe that this observation is not restricted to our case study. Secondly, we observe that for this case study the biggest emission reduction comes from shifting from road to rail transport. This finding is not generalizable because location to rail terminals and available capacity of the rail network may limit the applicability of intermodal rail transport in other cases. Nevertheless, it holds in general that emission reductions will be obtained by switching from air to road, road to rail, road to water or short sea, and rail to water or short sea; see, e.g., Defra (2010).

5.3. Impact of Price Sensitivity

We next examine the impact of price sensitivity on the profit reduction for a certain emission reduction target. This requires the estimation of demand parameters for all products (ϵ_j and Q_j). For each product, the actual demand is already given in the case study. We set the demand parameters such that the demand corresponds with the actual demand when $\lambda = 0$, the case without an emission constraint. Given a value of ϵ_j, Q_j can be calculated with the following formula:

$$Q_j = 2\tilde{q}_j + \epsilon_j(k_j + u_{1,j}),$$

where $\tilde{q}_j$ is the actual demand and $u_{1,j}$ is used since mode 1 is always chosen when $\lambda = 0$. Unless stated otherwise, the value of ϵ_j is the same for all products, and we remove the subscript j. In a linear demand function, the demand elasticity varies as a function of $p_{i,j}$; hence, the same value of ϵ may refer to different elasticity values across products.

First, we consider the situation that the price sensitivity is equal among all products and consider three different values: $\epsilon \in \{0.01, 1, 10\}$. We believe that this range of ϵ covers a broad range of realistic values for our data set. We also determine the solutions for the *item approach* for $\epsilon = 1$, in which an emission constraint is set per product instead of per group of products. We vary the value of the individual targets to obtain the set of solutions. In Figure 5, we present the relative profit, as a function of the emission reduction, for these scenarios.

The absolute value of the profit for a product and mode, and as a consequence the total profit, is decreasing in ϵ. When demand is less sensitive to price, higher prices can be charged. From Figure 5, it can be seen that a diminishing rate of return applies as described in §4.2. For a 1% profit reduction, 30%–38% emission reductions can be realized, depending on the value of ϵ, compared to 16% emission reduction when the item approach is used.

For a given emission reduction target, the profit reduction of the item approach versus the aggregate approach for $\epsilon = 1$ differs up to 21% (relative to the maximum profit without emission reduction), which is substantial. The difference is already substantial, but might be even higher when the unit cost of the products (k_j) is more diverse across products and when the total logistics cost is higher compared to the unit cost. We also observe that the larger ϵ, the larger

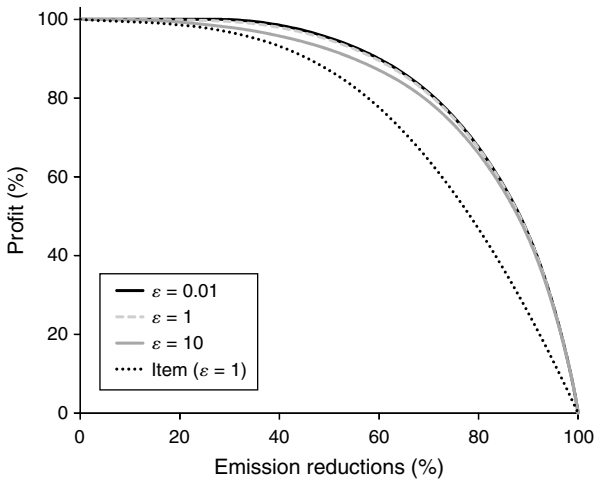


Figure 5 Compares the Optimal Solutions for the Item Approach and Aggregate Approach with Similar Elasticities

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the profit reduction for a given value of the total emissions, because a price increase leads to a relatively larger demand decrease (even though the differences are small).

We also consider two instances in which ϵ_j is nonhomogeneous across products. We divide the 56 products into three groups: low demand (18), medium demand (19), and high demand (19), and assign the following values $\epsilon_j = 0.01$, 1, or 3 to each group, respectively, which is denoted by *High demand, highly elastic* in Figure 6. In the second instance, $\epsilon_j = 3$ is assigned to lanes with low demand and $\epsilon_j = 0.01$ to lanes with high demand, which is denoted by *Low demand, highly elastic*. In the first case, the majority of the business value is represented by price-elastic demand, and in the second case the majority is represented by price-inelastic demand. As a result, the total profit (in absolute terms) is much higher in the first case.

If the price elasticity is nonhomogeneous, then the absolute value of the total profit is smaller than when $\epsilon = 0.01$ and larger than when $\epsilon = 10$. When the price elasticity is nonhomogeneous, the profit reduction is smaller than for the case $\epsilon_j = 0.01$, for a given value of Γ. This is explained by the fact that the profit of the high-demand products is smaller compared to e.g., the situation in which $\epsilon = 0.01$. Emission reductions are first realized by reducing emissions for high-demand products, which leads to large emission reductions for relatively smaller profit reductions. For the *Low demand, highly elastic* case, the high-demand products dominate the solutions and, as a consequence, it resembles the $\epsilon = 0.01$ graph.

For different levels of ϵ, the set of preferred modes per product changes, since the threshold values of the conditions change. When high-volume demand is price inelastic, the company can achieve emission reductions with less profit loss by switching modes and adjusting prices of high-volume product accordingly.

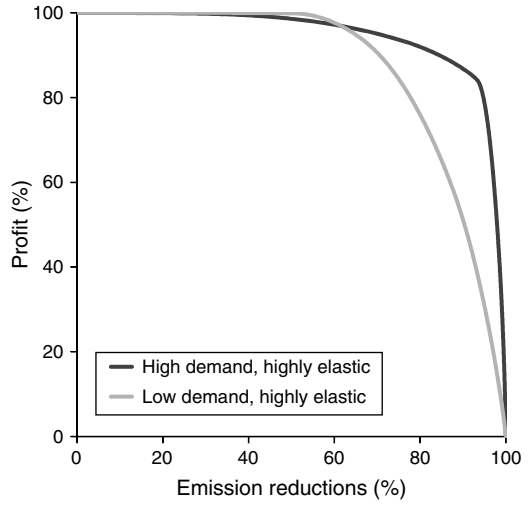


Figure 6 **Solutions for Scenarios with Different Demand Elasticities Between Product Groups (Low, Medium, and High Demand)**

From this extended case study, we generate the following insights: First, large emissions reductions can be achieved at relatively small profit losses. Second, whereas the required profit reduction to obtain a certain emission reduction target is determined by the price sensitivity of customers, the solutions are relatively robust to the price elasticity. Finally, we find that the portfolio effect can achieve emissions reduction of at most 21% higher profits (compared to the maximum profit in the situation without emission reduction) than an item approach.

6. Conclusion

In this study, we have considered a producer, who has outsourced all transport activities to a 3PL, wanting to determine the profit-maximizing transport mode allocation and sales price for each product, such that an overall emission target is met. An overall emission target allows for taking advantage of the portfolio effect, i.e., the emissions are reduced where it is cheapest, which is also the idea behind an emission trading scheme.

Lagrangian relaxation is employed to solve the problem, which is separable in the products. As a result, our model also enables us to estimate what emission-cost levels are required to reach certain emission reduction targets. The pricing decision can be solved separately and used as input to the transport mode selection decision. We have shown that the optimal price is linear in the total emissions and the profit is quadratic in the total emissions, for a given mode and product. This observation extends to the multiproduct setting. It can be concluded that a diminishing rate of return applies; i.e., emission reductions become increasingly expensive.

We have applied our method to a real-life case study, and we found that the transport emissions can be reduced by as much as 27%. The first 10% of emission reductions requires a 0.7% increase in the total logistics costs. This corresponds with a marginal cost of €100/tonne of CO_2. In the profit-maximization extension of the case study, we found that the emissions can be reduced by 30% for at most a 1.2% profit reduction, which does not appear to be sensitive to different price-elasticity scenarios. The value of allocating the emission target to individual products in such a way that the portfolio effect is exploited rather than using the same target for individual products is very significant. For example, an emission reduction of 50% results in a profit loss of 5% using the portfolio effect, whereas the profit loss for the same reduction without using portfolio effect is 13% under the same price sensitivity.

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For this data set, the emission reductions are mainly achieved by switching from road to rail transport, because of the characteristics of the European network and the problem environment. The general belief about transportation emissions is that the cheaper a mode is, the less carbon it emits. Although this intuition holds for single-mode transport in general, it is not necessarily the case for intermodal transport. In particular, our case study shows that intermodal transport is often more expensive than road transport because of longer lead times and additional handling costs associated with intermodal transport, but it results in low emissions. Hence, this demonstrates that maximizing profit does not necessarily result in minimizing emissions.

The emission reductions in the case study are achieved for relatively small profit reductions (or cost increase). In particular, a 10% emission reduction at only a 0.7% cost increase in the case study is a significant reduction given the fact that we only consider lanes within Europe (maximum distance 3,300 km) and that road transport is currently used. If the method is applied to a large-scale case study with intercontinental transport, one can expect larger emission reductions, because switching from air to ocean freight results in an extremely substantial emission reduction. Note that for intercontinental transport, the less carbon-emitting transport options (ocean or rail) have a higher share in the total transport because the first and last leg will only be a small part of the total distance. Nevertheless, this comes at the cost of increased lead times, and furthermore, ocean freight is not necessarily less costly for expensive items, considering the pipeline inventory costs.

Finally, we conclude that switching transport modes is an effective measure to reduce carbon emissions from transport, especially for small emission reduction targets, e.g., up to 20%. To reduce emissions even further (given the same infrastructure), an integrated approach that considers interactions with the 3PL and other producers is more efficient than switching transport modes alone. Possible such means are, e.g., collaboration with other producers to decrease empty returns and increase load factors, and sharing stock points.

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Appendix A

A.1. Alternative Model Formulation

An alternative model description for problem (P) can be obtained by applying the transformation $t_{i,j} = x_{i,j}p_j$ for the decision variables, which results in the following objective function:

$$x_{i,j}\Pi_{i,j}(p_j) = -\epsilon_j t_{i,j}^2 + (\epsilon_j(u_{i,j}+k_j)+Q_j)t_{i,j}x_{i,j} - (u_{i,j}+k_j)Q_j x_{i,j}^2$$

and emission constraint

$$x_{i,j}\Gamma_{i,j}(p_j) = e_{i,j}(Q_j x_{i,j} - \epsilon_j t_{i,j}).$$

The resulting model (P′) then has a quadratic objective function and linear constraints:

$$\begin{aligned}
(\mathrm{P}') \qquad \max \quad & \sum_{j\in J}\sum_{i\in I} x_{i,j}\Pi_{i,j}(p_j) \\
\text{subject to} \quad & \sum_{j\in J}\sum_{i\in I} e_{i,j}(Q_j x_{i,j} - \epsilon_j t_{i,j}) \le \beta, \\
& t_{i,j} \le p_j, \quad \forall\, i\in I,\ j\in J, \\
& t_{i,j} \ge p_j + M(x_{i,j}-1), \quad \forall\, i\in I,\ j\in J, \\
& \sum_{i\in I} x_{i,j} = 1, \quad \forall\, j\in J, \\
& x_{i,j}\in\{0,1\}, \quad \forall\, i\in I,\ j\in J, \\
& t_{i,j} \ge 0, \quad \forall\, i\in I,\ j\in J.
\end{aligned}$$

A.2. Proofs

Proposition 1. *Consider product $j\in J$ and $\lambda\ge 0$. Mode y maximizes the decentralized Lagrangian, $L_j^*(\lambda) = L_{y,j}^*(\lambda)$ and $x_{y,j}=1$ for $y = \arg\min(z_{i,j}(\lambda)\colon i\in I)$.*

Proof. To show: if mode $y\in I$ minimizes $z_{i,j}(\lambda)$ for $\lambda\ge 0$, then $L_{y,j}^* = \max\{L_{i,j}^*(\lambda)\colon i\in I\}$. Variable $z_{i,j}(\lambda)$ is the only mode-specific parameter in $L_{i,j}^*(\lambda)$. This implies that if $z_{i_1,j}(\lambda) = z_{i_2,j}(\lambda)$, then $L_{i_1,j}^* = L_{i_2,j}^*(\lambda)$ for $i_1, i_2\in I$. It remains to show that $L_{i,j}^*(\lambda)$ is nonincreasing in $z_{i,j}(\lambda)$. The function $L_{i,j}^*(\lambda)$ consists of two parts, depending on the value of $z_{i,j}(\lambda)$. For $z_{i,j}(\lambda)\le \hat{D}_j/\epsilon_j - k_j$, the first-order derivative of $L_{i,j}^*(\lambda)$ with respect to $z_{i,j}(\lambda)$ is:

$$\frac{\partial L_{i,j}^*(\lambda)}{\partial z_{i,j}(\lambda)} = \frac{1}{2}(\epsilon_j(z_{i,j}(\lambda)+k_j) - Q_j).$$

It follows directly that $L_{i,j}^*(\lambda)$ is decreasing in $z_{i,j}(\lambda)$ if: $z_{i,j}(\lambda) < Q_j/\epsilon_j - k_j$. For $z_{i,j}(\lambda)\ge Q_j/\epsilon_j - k_j$, $L_{i,j}^*(\lambda)$ is constant (at 0). Hence, $L_{i,j}^*(\lambda)$ is nonincreasing in $z_{i,j}(\lambda)$. □

Theorem 1. *Consider $i_2\in I$ and any $i_1, i_3\in I$ such that $u_{i_1,j} < u_{i_2,j} < u_{i_3,j}$ and $e_{i_1,j} > e_{i_2,j} > e_{i_3,j}$. Transport mode i_2 is selected if it meets condition* (a) *and* (b) *for any $i_1, i_3\in I$:*
Condition (a) $e_{i_2,j} \le \tilde{e}_{i_1,i_2}^j$,
Condition (b) $e_{i_2,j} \le \bar{e}_{i_2}^j(i_1, i_3)$,
where

$$\tilde{e}_{i_1,i_2}^j = e_{i_1,j}\frac{\hat{D}_j/\epsilon_j - k_j - u_{i_2,j}}{\hat{D}_j/\epsilon_j - k_j - u_{i_1,j}},$$

$$\bar{e}_{i_2}^j(i_1,i_3) = e_{i_1,j} + (e_{i_3,j} - e_{i_1,j})\frac{u_{i_1,j} - u_{i_2,j}}{u_{i_1,j} - u_{i_3,j}}.$$

Proof. *Proof of Condition a.* Since $u_{i_1,j} < u_{i_2,j}$ and $e_{i_1,j} > e_{i_2,j}$, $\lambda_{i_1,i_2}^j > 0$. Let $\hat{\lambda}_{i,j}$ be defined as the smallest positive value of λ such that $q_{i,J}^* = 0$: $\hat{\lambda}_{i,j} = 1/(e_{i,j})(\hat{D}_j/\epsilon_j - k_j - u_{i,j})$.

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Showing that $\lambda^j_{i_1,i_2} \le \hat{\lambda}_{i_1,j} \le \hat{\lambda}_{i_2,j}$ implies that mode i_2 is preferred over mode i_1 for a range of λ for which $L^*_{i,j} > 0$.

Let $e_{i_2,j} = \tilde{e}^j_{i_1,i_2} - \zeta$, where $\zeta \ge 0$. Let us first show that $\lambda^j_{i_1,i_2} \le \hat{\lambda}_{i_1,j}$:

$$\begin{aligned}\lambda^j_{i_1,i_2} &= \frac{u_{i_1,j}-u_{i_2,j}}{e_{i_2,j}-e_{i_1,j}}\\ &= \frac{u_{i_1,j}-u_{i_2,j}}{e_{i_1,j}((\hat{D}_j/\epsilon_j-k_j-u_{i_2,j})/(\hat{D}_j/\epsilon_j-k_j-u_{i_1,j}))-\zeta-e_{i_1,j}}\\ &= \frac{u_{i_1,j}-u_{i_2,j}}{e_{i_1,j}((u_{i_1,j}-u_{i_2,j})/(\hat{D}_j/\epsilon_j-k_j-u_{i_1,j}))-\zeta}\\ &\le \frac{u_{i_1,j}-u_{i_2,j}}{e_{i_1,j}((u_{i_1,j}-u_{i_2,j})/(\hat{D}_j/\epsilon_j-k_j-u_{i_1,j}))} = \hat{\lambda}_{i_1,j},\end{aligned}$$

where the inequality follows since the nominator and the denominator are negative.

Next, we show that $\hat{\lambda}_{i_1,j} \le \hat{\lambda}_{i_2,j}$:

$$\hat{\lambda}_{i_2,j} = \frac{1}{\tilde{e}^j_{i_1,i_2}-\zeta}\left(\frac{\hat{D}_j}{\epsilon_j}-k_j-u_{i_2,j}\right) \ge \frac{1}{\tilde{e}^j_{i_1,i_2}}\left(\frac{\hat{D}_j}{\epsilon_j}-k_j-u_{i_2,j}\right) = \hat{\lambda}_{i_1,j},$$

where the inequality follows since $\hat{D}_j/\epsilon_j - k_j - u_{i,j} > 0$, and the last equality follows from the definition of $\tilde{e}^j_{i_1,i_2}$.

Hence, $\lambda^j_{i_1,i_2} \le \hat{\lambda}_{i_1,j} \le \hat{\lambda}_{i_2,j}$.

Proof of Condition (b). If mode i_2 minimizes $z_{ij}(\lambda)$ for a certain value of λ, then it is preferred (follows from Proposition 1). This implies that for a given pair i_1, i_3, $u_{i_2,j} + \lambda e_{i_2,j} \le \min\{u_{i_1,j}+\lambda e_{i_1,j},\ u_{i_3,j}+\lambda e_{i_3,j}\}$ for at least one value of λ. This is shown next.

Assume that $e_{i_2,j} \le \bar{e}^j_{i_2}(i_1,i_3)$.

We first show that $\lambda^j_{i_1,i_2} \le \lambda^j_{i_1,i_3}$:

$$\begin{aligned}-\lambda^j_{i_1,i_2} &= \frac{u_{i_2,j}-u_{i_1,j}}{e_{i_2,j}-e_{i_1,j}} \ge \frac{u_{i_2,j}-u_{i_1,j}}{\bar{e}^j_{i_2}(i_1,i_3)-e_{i_1,j}}\\ &= \frac{u_{i_2,j}-u_{i_1,j}}{e_{i_1,j}+(e_{i_3,j}-e_{i_1,j})((u_{i_1,j}-u_{i_2,j})/(u_{i_1,j}-u_{i_3,j}))-e_{i_1,j}}\\ &= \frac{u_{i_3,j}-u_{i_1,j}}{e_{i_3,j}-e_{i_1,j}} = -\lambda^j_{i_1,i_3},\end{aligned}$$

where the inequality follows since $e_{i_2,j} - e_{i_1,j} \ge 0$.

Next, we show that $\lambda^j_{i_2,i_3} \ge \lambda^j_{i_1,i_3}$:

$$\begin{aligned}\lambda^j_{i_2,i_3} &= \frac{u_{i_2,j}-u_{i_3,j}}{e_{i_3,j}-e_{i_2,j}} \ge \frac{u_{i_2,j}-u_{i_3,j}}{e_{i_3,j}-\bar{e}^j_{i_2}(i_1,i_3)}\\ &= \frac{u_{i_2,j}-u_{i_3,j}}{e_{i_3,j}-e_{i_1,j}-(e_{i_3,j}-e_{i_1,j})((u_{i_1,j}-u_{i_2,j})/(u_{i_1,j}-u_{i_3,j}))}\\ &= \frac{u_{i_1,j}-u_{i_3,j}}{e_{i_3,j}-e_{i_1,j}} = \lambda^j_{i_1,i_3}.\end{aligned}$$

Therefore, we find the following ordering $\lambda^j_{i_1,i_2} \le \lambda^j_{i_1,i_3} \le \lambda^j_{i_2,i_3}$, which determines the ordering of the profits for any value of λ:

$$\begin{aligned}&u_{i_1,j}+\lambda e_{i_1,j} \le u_{i_2,j}+\lambda e_{i_2,j} \le u_{i_3,j}+\lambda e_{i_3,j} \quad \text{for } 0 \le \lambda \le \lambda^j_{i_1,i_2},\\ &u_{i_2,j}+\lambda e_{i_2,j} \le u_{i_1,j}+\lambda e_{i_1,j} \le u_{i_3,j}+\lambda e_{i_3,j} \quad \text{for } \lambda^j_{i_1,i_2} \le \lambda \le \lambda^j_{i_1,i_3},\\ &u_{i_2,j}+\lambda e_{i_2,j} \le u_{i_3,j}+\lambda e_{i_3,j} \le u_{i_1,j}+\lambda e_{i_1,j} \quad \text{for } \lambda^j_{i_1,i_3} \le \lambda \le \lambda^j_{i_2,i_3},\\ &u_{i_3,j}+\lambda e_{i_3,j} \le u_{i_2,j}+\lambda e_{i_2,j} \le u_{i_1,j}+\lambda e_{i_1,j} \quad \text{for } \lambda \ge \lambda^j_{i_2,i_3}.\end{aligned}$$

Recall $u_{i_1,j}+\lambda e_{i_1,j} \le u_{i_3,j}+\lambda e_{i_3,j}$ for $\lambda \le \lambda^j_{i_1,i_3}$ and $u_{i_1,j}+\lambda e_{i_1,j} \ge u_{i_3,j}+\lambda e_{i_3,j}$ for $\lambda \ge \lambda^j_{i_1,i_3}$. From this, we can conclude that $u_{i_2,j}+\lambda e_{i_2,j} \le \min\{u_{i_1,j}+\lambda e_{i_1,j},\ u_{i_3,j}+\lambda e_{i_3,j}\}$ for at least one value of λ. □

A.3. Details of Emission Calculation

The general required parameters and the source (assumption or actual data) to calculate the emissions from transport with the NTM model in the Cargill case are:

Transport mode. Given by LSP and Cargill, road or ferry by default, intermodal rail, water, or short sea for the intermodal bids.

Distance per mode. Calculated based on origin, destination, and transshipment locations.

Weight and volume of product. Given by Cargill, weight is restricting factor.

Payload. Given by the LSP.

TEU (*twenty-foot equivalent unit*). Determined by product density and capacity of equipment (1 or 1.5 TEU).

Cleaning. Steam cleaning is assumed for all shipments.

Additionally, we specify the following mode-specific assumptions.

Road transport. All assumptions are taken from NTM Road (2008) unless indicated otherwise. The formula to calculate the unit emissions for road transport is:

$$e_{i,j} = B_i \delta_{i,j} \frac{w_j}{\tilde{w}_i}.$$

We assume that a truck semitrailer is used for all road transport for which the maximum load is 40 tonne. The load factor is determined by the payload. Moreover, in line with NTM, we estimate a 20% positioning distance to be added to the travel distance. Further, we assume no empty returns.

The emission factor is determined by the fuel consumption (in L/km) and the fuel emissions (in kg/L). The fuel consumption depends on the road type, and the values are given in Table A.1. In our calculations we have assumed 85% highway, 10% rural and 5% urban. One liter of diesel produces 2.6 kg of CO_2.

Rail transport. Two types of rail transport are distinguished: electrical and diesel. All assumptions in this section are taken from NTM Rail (2008) unless indicated otherwise. The emissions for one unit shipped by rail transport are determined with the following equation:

$$e_{i,j} = \sum_{z \in Z} \epsilon_{i,j}(z) \delta_{i,j} w_j,$$

where the emission factors for electrical ($\epsilon^1_{i,j}(z)$) and diesel rail ($\epsilon^2_{i,j}(z)$) in country z are:

$$\epsilon^1_{i,j}(z) = \frac{T(\tau^z)\cdot EE^z}{10^6\sqrt{W_{gr}}\cdot\sigma_{i,1}\cdot(1-\eta)}, \quad \epsilon^2_{i,j}(z) = \frac{T(\tau^z)\cdot FE^z}{10^6\sqrt{W_{gr}}\cdot\sigma_{i,2}}. \quad (14)$$

Table A.1 Fuel Consumption

Road type	Motorway		Rural		Urban	
Load factor (%)	0	100	0	100	0	100
Fuel consumption	0.226	0.360	0.230	0.396	0.288	0.504

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For rail transport, the distance is divided into distances per country because the emissions to generate 1 kWh, and the fuel content, differ per country. Define the set of countries through which the train travels as Z and $z \in Z$. In our calculations, we do not distinguish between different countries but assume Europe averages. We also use the overall shares of electrical and diesel rail: 76.6% and 23.4%, respectively.

The emission factor for rail depends on the gross weight of the train W_{gr}, an emission constant, corrected for the terrain $T(\tau^z)$, the load factor $\sigma_{i,1}$, energy (or fuel) efficiency factor EE^z and FE^z, and a transfer loss (for electrical only) η. Per country the terrain is specified as flat, hilly or mountainous. We use a 72% load factor, $\sigma_{i,1} = 0.72$. We assume that the gross weight of the train equals 1,000 tonne, $W_{gr} =$ 1,000 tonne. $\eta = 0.10$. For diesel rail, we assume that $FE =$ 3.175 kg CO_2/kg diesel.

Water transport. Water transport covers short-sea transport and inland transport with diesel oil-powered vessels. All details are taken from NTM Water (2008). The formula to calculate the unit emissions for road transport is:

$$e_{i,j} = \beta_{i,j} \cdot FC_i \cdot FE_i \cdot \delta_{i,j},$$

where $\beta_4 =$ unit capacity/total capacity. We assume a bulk water-type feeder for short sea transport, and for inland water default NTM values for a container vessel. We assume short-sea load factor of 80%. For inland water, use default NTM value. For water transport, burning 1 kg of diesel results in 3.2 kg of CO_2.

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